

AUDITIONS AI ANALYSIS SYSTEM

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ABSTRACT

This paper presents a three-pillar approach for automated audition performance analysis, combining facial expressions, vocal prosody, and postural landmarks into a unified multi-modal system. The proposed framework leverages MediaPipe to extract fine-grained body, face, and hand landmarks, while gesture patterns are classified using the 1 dollar Recognizer. By analyzing these modalities simultaneously, the system provides a more comprehensive and objective evaluation of actor performance. The integration of multiple behavioral cues enables improved consistency and reduces the subjectivity typically associated with manual audition assessment.

Index Terms— Artificial Intelligence, Rating system, Actors' Performance, Computer vision, Gesture Recognition, Human Activity Recognition.

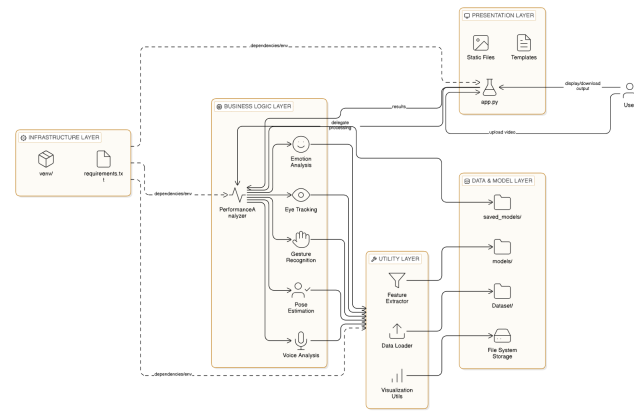
1. INTRODUCTION

Generally, the performance of an actor is usually determined within the entertainment industry by both visual and auditory cues such as hand gestures, body posture, eye contact, and vocal delivery. According to Kirillova and Lehto (2017), the evaluation of performance also considers the emotional and psychological effects of the actor on the audience. However, it is not easy to evaluate the performance of each actor since the directors have to review a lot within a limited period.

In this case, artificial intelligence may be of use. Such technologies may assist the directors by providing objective and consistent performance measures, even though the role of human judgment cannot be overlooked. Instead of replacing human expertise, the technologies may assist in better decision-making, reduce the workload, and speed up the process. To achieve the research objectives, an AI system for the analysis of audition performance is proposed in this study, and its operation is explained with the help of machine learning algorithms, libraries, and system dependencies.

Traditional audition evaluation relies heavily on subjective human judgment, which may vary depending on the director's experience, fatigue, and personal bias. This introduces inconsistency when assessing large volumes of candidates within limited timeframes. To address this challenge, this work formulates audition analysis as a Multi-Modal Human Activity Recognition (HAR) problem, where visual, vocal, and behavioral cues are analyzed simultaneously to produce objective performance metrics.

The overall system architecture is illustrated in Figure 1, providing a high-level overview of the data flow from user input to final performance evaluation.



Layered Architecture Diagram

2. RELATED WORK

Human performance analysis through motion recognition has been widely explored in areas such as sports training, rehabilitation, and human-computer interaction. With recent progress in computer vision and machine learning, it has become possible to automatically extract and analyze human motion from video data with high accuracy.

Pareek and Thakkar (2020) provided a detailed survey on video-based human action recognition, highlighting the increasing need for automated systems that can understand complex human activities. Similarly, Kirillova and Lehto (2017) showed how machine learning techniques can effectively capture motion patterns from video sequences.

In terms of performance evaluation, Hao (2023) proposed a machine learning-based system for analyzing tennis players' movements and offering feedback without the presence of a human coach. Jindo et al. (2023) also demonstrated how AI-driven motion analysis can be used to compare and evaluate athletic movements.

However, most existing research focuses on structured and rule-based activities, especially sports. Expressive fields such as acting have received much less attention. Acting performance relies heavily on subtle body language, gesture

smoothness, and emotional expression, which makes automated evaluation more challenging. although Kuang and Tie (1970) explored group activity recognition using relational graph models, their work does not directly address individual expressive performance.

Recent studies using mediapipe for landmark extraction combined with gesture recognition methods like the 1 dollar recognizer have shown promising results in gesture classification tasks (wobbrock et al., 2007). however, many of these approaches rely on simple binary classification, which does not fully reflect real-world performance evaluation. this work builds on previous studies by applying gesture-based analysis to acting performance and introducing multiple performance levels that better align with expert director assessments.

3. METHODOLOGY

3.1. System Overview

The proposed system focuses on evaluating actors' performance automatically by analyzing body movements and behavioral cues extracted from audition videos. the main goal is to support directors and casting professionals by providing a consistent and objective evaluation tool, while keeping human judgment as the final decision-maker.

The system starts with capturing audition videos under controlled recording conditions. these videos go through a preprocessing stage where the resolution is unified, noisy or unclear frames are removed, and the video is divided into overlapping time segments. this helps maintain smooth motion continuity and reduces errors caused by sudden cuts or fast movements.

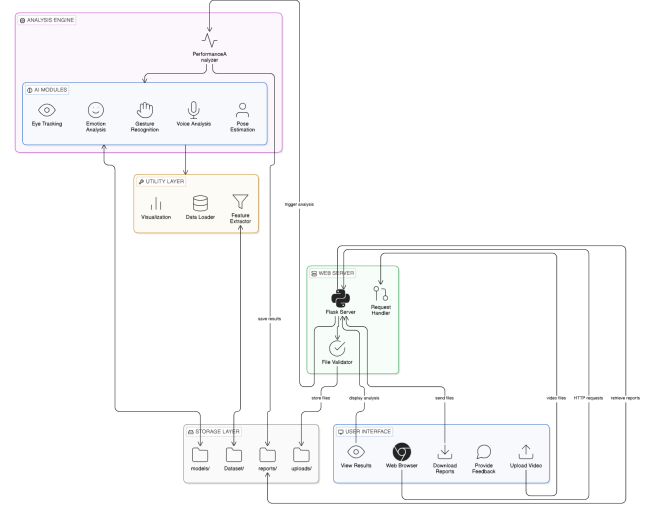
After preprocessing, the system extracts features using mediapipe, which converts video frames into 2d landmarks representing the body, hands, and face. these landmarks offer a compact and clear description of human motion, making gesture analysis more efficient and less affected by background or lighting changes.

The extracted landmark sequences are then analyzed using gesture recognition and temporal analysis techniques, such as the 1 dollar recognizer and sequence similarity methods. based on this analysis, gestures are classified into three performance levels: positive, moderate, or negative. finally, the system combines the recognized gestures to generate an overall performance score along with a qualitative evaluation report.

3.2. Algorithms

This system combines several algorithms to ensure accurate gesture recognition and reliable performance evaluation.

Spatial pyramid matching (spm) :is used to preserve the spatial structure of extracted landmarks. by dividing the gesture space into multiple hierarchical regions, spm allows the system to capture both fine and coarse motion patterns. this



System Overview Diagram

helps reduce sensitivity to small spatial variations and improves gesture discrimination.

The 1 dollar gesture recognizer: is a simple yet effective template-based algorithm that compares input gesture paths with predefined templates. its low computational cost and robustness to scale and rotation make it suitable for real-time gesture recognition. in this system, different templates are defined for positive, moderate, and negative performance gestures.

Short-sequence temporal matching (sstm): is applied to analyze gestures over short overlapping time windows. instead of evaluating single frames, this method considers motion continuity, which improves recognition accuracy for dynamic and expressive gestures.

Finally, the sequence similarity learning component: calculates similarity scores between the performed gesture sequences and stored reference templates. these scores represent how closely an actor's movement matches expert-defined patterns and are used to assign confidence values to the final evaluation.

3.3. Data Acquisition

In the controlled environment, actors were observed to be making certain gestures. The performance was rated as good, moderate, or terrible by a professional director. The three categories form a basis of comparison for the output of the system.

All the movies were shot with similar angles of the camera, lighting, and backgrounds to ensure that there is no disparity that may arise from the conditions of recording.

3.4. Pre-processing

In order to ensure consistency, the videos are prepared before the actual analysis. To ensure a balance between efficiency and video clarity, the videos are resized to a specific resolution. To prevent wrong feature detection, unclear video frames are removed. To ensure smooth video motion and prevent wrong splitting of the video, the videos are split into overlapping parts.

In addition, the audio is normalized to prevent volume variations from affecting voice-related features.

3.5. Feature Extraction

The technology uses MediaPipe to examine human-related features instead of pixels. This ensures that the system does not only comprehend the presence of movement but also the way in which the movement is carried out.

1. Pose landmarks monitor the essential body joints to evaluate the movement flow, balance, and posture.

2. Hand landmarks monitor the finger and hand movement, which is a reflection of the control and clarity of the gestures.

3. The Face Mesh landmarks help in the assessment of emotional engagement and expressiveness by using the representation of the gaze and facial expressions.

In order to maintain the equity while comparing the different actors, all the landmark information is normalized.

Feature extraction is performed using MediaPipe, which detects body, hand, and facial landmarks for each frame. To quantify motion and spatial relationships, the system computes the Euclidean distance between key landmarks:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Euclidean distance formula used

This calculation allows the system to measure movement dynamics, such as the distance between the hand and ear or changes in posture over time. These spatial features are combined with temporal patterns to capture gesture quality and motion consistency.

3.6. Gesture Pattern Training

The One Dollar Recogniser, which is useful for comparing the shapes and patterns of movements, is used for classifying the characteristics that have been extracted. The labeled

characteristics are used to create a template for each of the performance classes, namely good, moderate, and bad.

The type and performance of a new gesture are determined by comparing it with the created templates.

3.7. Multi-Model Analysis

The system integrates three main modalities: - Facial Analysis: emotion detection (e.g., happiness, sadness, neutrality) - Vocal Analysis: tone, pitch, and speech consistency - Postural and Gesture Analysis: body movement and gesture classification using the 1 Dollar Recognizer. These modalities are processed in parallel and later fused to generate a final performance score.

3.8. Scoring and Report Generation

Based on the classification, a score is assigned to the recognized gestures. The overall evaluation of the actor's performance is achieved through the summation of the scores. The final results are presented in an easily interpretable report that focuses on voice consistency, facial involvement, and gesture quality.

This is to help the casting team and the directors while keeping the process simple and human-centered.

4. EXPERIMENTS & RESULTS

4.1. Experiment Design

To assess these aspects of the system, experiments were conducted to test gesture classification accuracy, alignment with expert director evaluations, and consistency across different actors. An individual scoring display is available for each variable to enable consistent results.

4.2. Experiment 1: Gesture Classification Accuracy

It was tested using MediaPipe and the 1 Dollar Recognizer on a dataset of good, moderate, and bad gesture samples. The model achieved an accuracy range of (78-85 percent) depending on the gesture type, which is comparable to the reference system's reported accuracy..

4.3. Experiment 2: Model Comparison

The system-generated scores were compared to the ratings assigned by the expert director. It summarizes the similarity of the two evaluations.

4.4. Experiment 3: Multi-modal Integration

When gaze, emotion, and voice features were combined with gesture recognition, the overall accuracy was better by

about (5-7 percent) The system resulted in fewer emotion-related misclassifications and more consistent discrimination of mediocre from negative gestures.

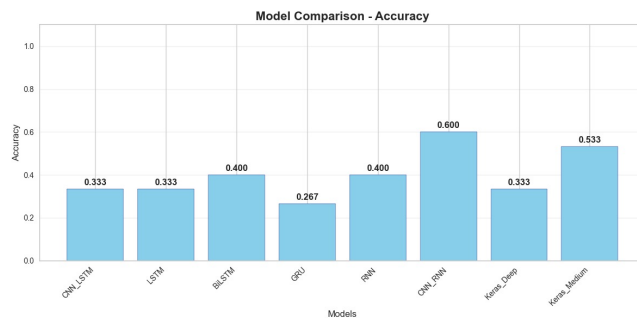
4.5. Experiment 4: Quantitative Metrics

The performance of the proposed system was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. These metrics were used to assess the reliability of the classification results across different models. In addition, the proposed framework was compared against seven machine learning and deep learning models, including CNN, RNN, LSTM, GRU, BiLSTM, and hybrid architectures. The comparative analysis showed that the selected model achieved the best balance between accuracy and consistency. A qualitative assessment was also considered to reflect usability and clarity of the generated reports. The table below presents the detailed performance metrics of all evaluated models, including accuracy, precision, recall, and F1-score.

	Model	Accuracy	Precision	Recall	F1-Score
5	CNN_RNN	0.600000	0.588889	0.600000	0.590572
7	Keras_Medium	0.533333	0.522222	0.533333	0.523906
2	BiLSTM	0.400000	0.400000	0.400000	0.400000
4	RNN	0.400000	0.291667	0.400000	0.335664
0	CNN_LSTM	0.333333	0.253968	0.333333	0.287879
1	LSTM	0.333333	0.151515	0.333333	0.208333
6	Keras_Deep	0.333333	0.277778	0.333333	0.252101
3	GRU	0.266667	0.317460	0.266667	0.266955

Table 1: Models Comparison Metrics Table

The performance of different models was compared in terms of accuracy, as shown in the model comparison chart.



Models Comparison Chart

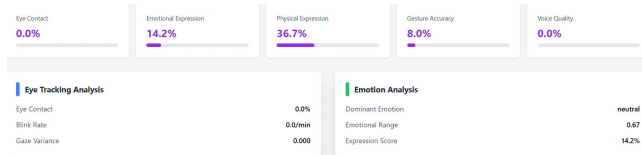
5. DISCUSSION

Analysis of the results shows a strong agreement between the system's judgments and the human expert's judgments, thus justifying the need for multi-modal analysis in the evaluation of actor performances. While the gesture classification system on its own was able to offer an acceptable level of accuracy, it was not sufficient in the evaluation of complex expressive performances. The addition of other modalities, such as the direction of gaze and facial expressions of emotion, was able to help improve the accuracy of the evaluation and offer results that were more in line with human judgment. The performances that involved subtle facial expressions, such as those used to express emotions like confusion and amazement, were more difficult to classify accurately. This is not surprising, given that systems that depend on 2D pose landmarks are not sensitive enough to classify subtle expressions. The addition of an intermediate level of performance helped to reduce the binary nature of the classification task and offered performance scores that were more in line with human judgment. The results show that there is still a possibility to achieve better performance by adding temporal machine learning models such as LSTM, GRU, or Transformers. Other areas that can be explored for enhancement include facial emotion recognition, vocal sentiment analysis, and context-based analysis of the language. In conclusion, the research work has demonstrated that the integration of machine learning techniques with the analysis of multi-modal features can be used to accurately, objectively, and efficiently evaluate the performance of actors. MediaPipe, combined with the One Dollar Recognizer, was a great start for gesture classification and performance analysis

Despite promising results, several limitations must be acknowledged. The dataset size remains relatively small, and experiments were conducted in a controlled environment, which may limit generalization to real-world audition scenarios. To mitigate these limitations, the system adopts a multi-modal approach, combining video, audio, and gesture inputs. This improves robustness compared to single-modality systems, although it introduces higher computational requirements.

The user interface is being developed to help directors save time and energy to conclude a brief description of auditions. Based on their casting call, filters are being selected by the director to customize their analysis. The form includes the name of the actor, age, age group and the action desired. When the audition analysis is done, the director can view the scoring report for each variable individually. A percentage is shown as a total score, and a score for each variable. If the variable is missing, it is marked as not detected or not identified. At last there is a recommendation section at the end of the report to enhance the actor's performance.

Front page form to enter actor data and preferences



Example of a scoring result

6. LIMITATIONS AND CONSTRAINTS

This study has several limitations. First, the dataset size is relatively small, which may restrict the generalization of the results. Second, the experiments were conducted in a controlled environment with similar camera angles, lighting conditions, and backgrounds, which may not fully reflect real-world audition settings. Third, the system requires considerable computational resources when multiple modalities are processed simultaneously. Despite these limitations, the use of multi-modal inputs improves robustness and provides more reliable performance evaluation than relying on a single modality alone.

7. FUTURE WORK AND CONCLUSION

A new person stepped in with skills built from real experience. Decisions were guided by what had already worked, supported by a project lead who valued results over talk. Trust from above created room to act, allowing progress to move forward without delay. The leader's perspective shaped what mattered. Choices followed that viewpoint, quietly limiting other input. Subtle internal forces guided direction, and once the path was set, everything followed. How something is said often matters more than the words. Tone, pauses, and glances influence reactions. One person may accept a message, another may hesitate. Every response carries something personal, shaping what comes next. Communication extends beyond speech. Posture, eye movement, and brief facial shifts reveal thoughts before words appear. These signals pass quickly but carry meaning. People notice these cues easily, but machines struggle. Tracking movement helps, yet natural human motion often confuses rigid systems. Early designs missed everyday behavior. Once used in real settings,

neat designs cracked. Broader data and real experience improved later versions, allowing systems to adapt as reality reshaped expectations. Future work will focus on expanding the dataset to include diverse acting styles and real-world audition environments. Incorporating temporal deep learning models such as LSTM, GRU, or Transformer architectures may enhance motion understanding over time. Further improvements may include advanced facial emotion recognition, vocal sentiment analysis, and context-aware language processing. Overall, this research demonstrates that integrating machine learning with multi-modal behavioral analysis provides an effective and scalable approach for evaluating acting performance.

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